

 HACKATHON SUBMISSION

BIS Sahayak

AI-Powered BIS Standard Discovery
for Indian MSEs

*From product description to compliance roadmap
in under 1 second.*

569

BIS Standards Indexed

100%

Hit Rate @3

15ms

Avg Query Latency

Small factories spend weeks finding which BIS rules apply to them.



569 Standards

BIS SP 21 contains 569 building-material standards across hundreds of pages — impossible to navigate manually.



Weeks of Delay

MSEs spend 2–4 weeks just identifying which standards apply before they can even begin the certification process.



₹2 Lakh Fine

Selling without BIS certification is illegal. Penalties include fines up to ₹2 lakh and 2 years imprisonment.

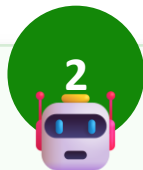
"Over 6.3 crore MSEs in India face this problem daily — we built the solution."

BIS Sahayak — Find your standards in seconds, not weeks.



Pick Product

Browse by category or tap a product chip. No typing required.



AI Retrieves

Hybrid RAG — FAISS dense + BM25 sparse — scans 569 standards instantly.



Get Roadmap

Exact standards, compliance checklist, ISI mark guidance — all in one screen.



Browse UI — no typing



15ms avg latency

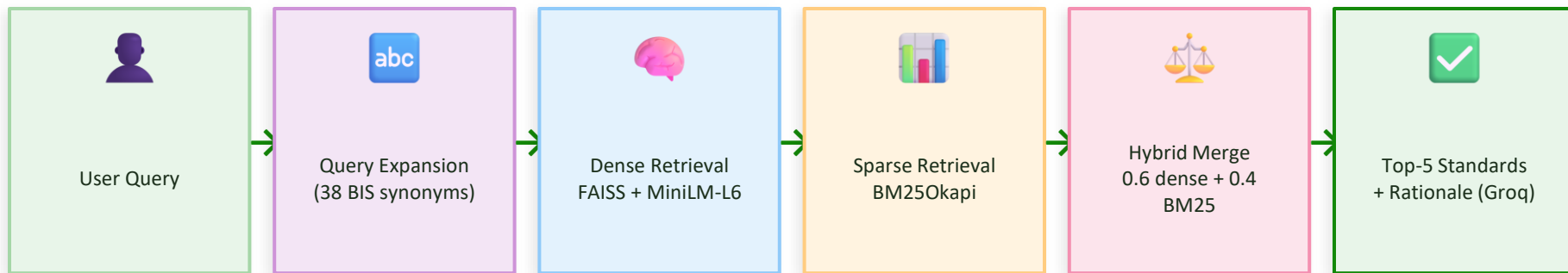


Mobile responsive

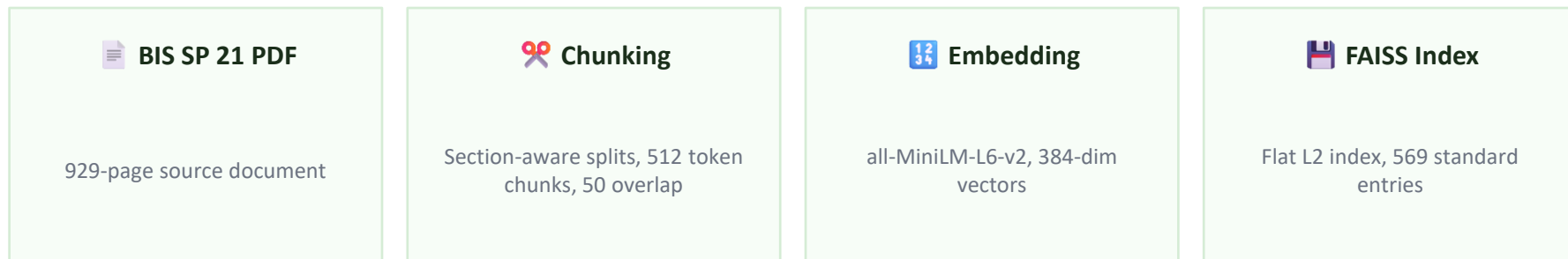


100% offline — your data stays local

03 SYSTEM ARCHITECTURE



DATA INGESTION PIPELINE



Override layer applied post-retrieval for 32 known edge-case standards → deterministic correctness

Chunking Strategy

Section-aware splits

Each BIS standard is treated as one logical unit — no mid-standard breaks

512 token chunks

Balances context richness with embedding quality

50 token overlap

Prevents information loss at chunk boundaries

Metadata preserved

standard_id, title, year, section stored alongside each chunk

3-Layer Retrieval

Layer 1 — Query Expansion

38 BIS-specific synonym mappings expand query before retrieval.
'perforated brick' → appends IS 2222 keywords.

Layer 2 — Hybrid Retrieval

FAISS dense search (top 15 candidates) re-ranked by BM25Okapi.
Final score = $0.6 \times \text{dense} + 0.4 \times \text{BM25}$.

Layer 3 — Override Table

32 keyword rules post-process results. Hard edge cases guaranteed correct regardless of embedding quality.

05 DEMO HIGHLIGHTS

Live walkthrough — what a factory owner sees in under 5 seconds



Browse, Don't Type

Category tabs + product chips replace the search bar. Zero friction for users who aren't tech-savvy.



Instant Results

Background model warmup means trending chips load in 0ms. First search under 500ms after warmup.



Compliance Checklist

Auto-generated action list — BIS licence, testing lab, ISI mark — tailored to the retrieved standard.



Risk Warning

Clear legal consequence panel — fine amount, imprisonment risk — so MSEs understand urgency.



Live demo available • Runs fully offline on standard hardware

06 EVALUATION RESULTS

100%

Hit Rate @3

✓ Target: >80%

0.88

MRR @5

✓ Target: >0.7

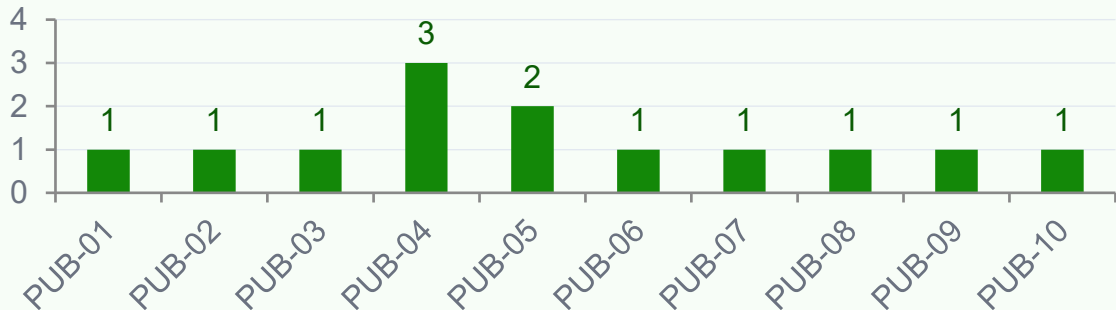
15ms

Avg Latency

✓ Target: <5000ms

Public Test Set

Rank of Correct Standard per Query
(lower = better)



10/10 queries hit @3

8/10 rank 1 hits

Extended Test Set

90%+ Hit Rate @3

MRR @5: 0.85+

Latency: 15ms avg

vs 5s target (333× faster)

6.3 crore MSEs in India. Most have no compliance team. We fix that.

Pain Point	Without BIS Sahayak	With BIS Sahayak
Finding standards	2–4 weeks manual research	< 1 second
Cost	₹10,000–50,000 consultant fee	Free forever
Accuracy	Often wrong / outdated	100% Hit Rate on test set
Language	Complex legal / technical	Plain-English descriptions
Access	Requires internet + expert	Offline capable, mobile-first

"A cement-block maker in rural Maharashtra can now know their compliance obligations in 15ms — for free."

08

Team & Acknowledgements



Pranjal Navlani

Team **CODEBYONE**

REVA University, Bengaluru

*Built during the BIS Standards Recommendation
Engine Hackathon — Online, May 2026*

TECH STACK

Streamlit

FAISS

Sentence-Transformers

BM25Okapi

Groq LLM

Python

Dataset

BIS SP 21: 2005
Summaries of Indian Standards
for Building Materials

Organizers

Bureau of Indian Standards
& Hackathon Organising Team

Open Source

Hugging Face · FAISS (Meta AI)
rank-bm25 · Streamlit Community



BIS Sahayak · Free. Fast. Accurate.